

Session 13: Prompting for Real Tasks

You've mastered the art of prompting - from basic instructions to advanced techniques like few-shot, role-based, and chain-of-thought. Now it's time to put your skills to work in the real world.

In this session, you'll use AI to tackle creative projects, everyday tasks, research, learning, writing, etc. You'll also learn how to use AI responsibly.



Recap : What We Learned in Sessions 11 & 12

Key Idea	What It Means
Prompt	Your instruction to AI.
Key Elements	Clear Instructions, Context, Format, Constraints – tools you can use when needed.
Common Mistakes	Assuming AI understands, incomplete info, accepting first answer, overloading, not checking output.
Iterative Prompting	Refining through multiple rounds.
The Prompt Framework	Action + Topic + Context + Format + Constraints.
Zero-shot	No examples - quick and simple.
Few-shot	Give examples to guide the AI.
Role-based (Persona)	Tell AI to act as a tutor, coach, or character.
Chain-of-thought	Ask AI to "think step by step."

The Most Important Idea from S11 & S12

There's no single perfect prompt - use helpful elements when needed, iterate, and always stay in control. Advanced techniques give you even more power to get useful, accurate, and creative results.

Today: We'll take these skills and apply them to **real tasks** - whatever you face in school, life, and creativity while keeping the "human-in-the-loop" principle front and center.

Iterate

Refine through multiple rounds

Stay in Control

You are always the human in the loop

Apply

Use skills on real tasks today

Your AI Toolkit for Real Tasks

You now have access to four powerful tools. Each has strengths:

Tool	Best For	Example
ChatGPT	Conversational AI, creative writing, general Q&A, image generation	"Write a short story about a robot who learns to paint."
Gemini	Multimodal tasks (text + images), deep research, long context	"Upload this diagram of a plant cell and explain each part."
NotebookLM	Source-grounded research – answers based only on documents you provide	"Upload your textbook chapter, then ask questions – no hallucinations."
QuillBot	Rewriting, paraphrasing, improving grammar and style	"Take this paragraph and make it more formal."

 **Key insight:** The right tool depends on your goal. For creative writing, ChatGPT or Gemini shine. For research with sources, NotebookLM is your best friend. For polishing your own writing, QuillBot is a lifesaver.

Objectives

01

Creative Tasks

Use advanced prompting for storytelling, art, game design, etc.

02

Everyday Tasks

Apply prompting to planning, organizing, decision-making, etc.

03

Research

Use AI for research while verifying facts and staying source-grounded

04

Learning

Create prompts that support math, science, history, languages, etc.

05

Writing Partner

Use AI to draft, revise, and polish

06

Responsible AI

Understand and practice responsible AI use - how to use AI ethically and avoid cheating

07

Prompt Library

Add real-task prompts to your My Prompt Library



Warm-Up : Real Tasks, Real Instructions

Think about something you've done recently - maybe wrote a story, planned a weekend trip, studied for a test, or wrote an email. How did you figure out what to do? Did you ask someone for help? Did you look up examples? Did you make a list?

Now imagine you could ask AI to help. How would you ask it to **spark ideas, organize your thoughts, or explain a tricky concept?**

📝 **Big Question:** How can AI be a **thinking partner** for the tasks you already do every day? Today we'll find out.

Prompting for Creative Tasks

AI can be a fantastic **creative partner** - helping you brainstorm, develop characters, imagine settings, or even generate names and dialogue. The key is to use it to **spark your own creativity**, not to replace it.



Prompting for Creative Tasks

Creative tasks might include:

- Writing stories, poems, or scripts
- Designing characters, worlds, or games
- Coming up with titles, names, or slogans
- Inventing recipes, products, or magical creatures
- Any task where imagination leads the way

How to prompt for creativity:

Use **personas**, **few-shot examples**, and **open-ended instructions** to get ideas that you can build on.

Creative Prompting - Examples

Goal	Prompt (with techniques)
Character for a story	"Act as a fantasy character creator . Give me three ideas for a friendly forest spirit who helps lost travelers. For each idea, include: name, appearance, and one magical ability. Use bullet points."
Plot twist	"I'm writing a mystery story set in a school. Here's an example of a plot twist: 'The missing trophy was hidden in the principal's office the whole time – and the principal knew it.' Now give me two more plot twists for a school mystery, each one surprising but believable."
Story starter	"Act as a storytelling coach . Give me three different opening sentences for a sci-fi story about a robot who discovers it can dream. Each opening should set a different mood: one mysterious, one funny, one hopeful."
Game idea	"I want to create a simple card game for two players. Think step by step about what makes a fun game: rules, win condition, and a theme. Then describe one game idea in three bullet points."

 **Key Insight:** AI gives you a **starting point**. You take the ideas, use your creativity and make them your own.



Prompting for Everyday Tasks

AI can help with the small (and big) tasks of daily life - planning, organizing, making decisions, and even remembering things. Use it to **lighten the mental load**, but **always make the final call yourself**.

Prompting for Everyday Tasks

Everyday tasks might include:

- Planning outings, trips, or schedules
- Creating to-do lists, packing lists, or shopping lists
- Making decisions (what to cook, what to wear, where to go)
- Organizing time for studying or projects
- Any practical task where organization helps

How to prompt for everyday tasks:

Use **clear instructions**, **constraints** (time, budget, preferences), and sometimes **personas** (e.g., "act as a travel planner") to get practical, actionable suggestions.

Everyday Prompting - Examples

Goal	Prompt (with techniques)
Weekend plan	"Plan a one-day trip to a nearby park for a family of four. We have a budget of \$50 and want to spend about 4 hours there. Include: what to bring, one activity for kids, and a suggested meal. Use bullet points . Keep it simple."
Packing list	"Here's an example of a packing list for a beach day: 'Sunscreen, towel, hat, water bottle.' Now make a packing list for a hiking trip in the mountains for one day. Include clothing, safety items, and snacks . Use bullet points."
Decision help	"I'm trying to decide between baking cookies or making a fruit salad for a school bake sale. Think step by step about time, cost, and how many people will eat it. Then give me a recommendation with your reasoning."
Study schedule	"Act as a study coach . I have a math test in 5 days and 2 hours each day to study. Create a simple study plan that covers the main topics: fractions, decimals, and word problems. Use a bulleted list with one task per day."

 **Key Insight:** AI helps you **organize**, but you decide what works for your life.

Mini Activity 1 :

Prompting for Everyday & Creative Tasks

Instructions: For each prompt below, decide whether it's for a **creative task** or an **everyday task**. Then write a quick improved version using at least one technique (few-shot, persona, chain-of-thought, etc.).

Prompt	Creative or Everyday?	Your Improved Prompt
"Give me ideas for a new superhero."		
"Plan a picnic for Saturday."		
"Write a funny poem about a cat."		
"Make a list of groceries for a taco dinner."		

Sample answers on next slide.

Mini Activity 1 : Answer Key

Prompt	Type	Improved Prompt (Example)
"Give me ideas for a new superhero."	Creative	"Act as a comic book creator. Give me three ideas for a superhero whose power comes from music. Include a name, costume detail, and one weakness."
"Plan a picnic for Saturday."	Everyday	"Plan a simple picnic for two people on Saturday afternoon. Include food ideas that don't need cooking, a suggested location (park or beach), and one game to play. Use bullet points."
"Write a funny poem about a cat."	Creative	"Here's an example of a funny poem about a dog: 'My dog thinks he's a kangaroo / He jumps on beds and couches too.' Now write a similar funny poem about a cat who loves to knock things off tables."
"Make a list of groceries for a taco dinner."	Everyday	"I'm making tacos for four people. List the ingredients I'll need, organized by store section: produce, meat, dairy, and pantry. Keep it simple."



Research with AI

AI can help you **find, summarize, and organize information** but it's not a replacement for real sources. Always **verify facts** and **use reliable sources** (like NotebookLM with your own articles or trusted sources).

Research with AI

Research tasks might include:

- Gathering facts for a project or report
- Summarizing articles, chapters, or videos
- Comparing different viewpoints or arguments
- Preparing for a debate, presentation, or discussion
- Any task where you need accurate, organized information

How to prompt for research:

Combine **NotebookLM** (with your uploaded sources) with **ChatGPT** or **Gemini** to find, summarize, organize, compare, etc. Always ask for **citations** or **source checks**.

Research Prompting - Examples

Goal	Prompt (with techniques)
Summarize an article	(In NotebookLM, after uploading an article) "Summarize this article in 3 bullet points for a 6th grader. Focus on the main idea and two key facts."
Compare viewpoints	(In ChatGPT or Gemini) "I'm researching whether zoos are good for animals. Give me two arguments for and two arguments against , each with a one-sentence explanation. Format as a table."
Find reliable sources	(In ChatGPT or Gemini) "I need three reliable sources about the ancient Maya civilization. For each, give the title, author (if known), and why it's trustworthy. Do not invent sources - if you're unsure, say so."
Check a fact	(In ChatGPT or Gemini) "I read that the Great Wall of China is visible from space. Is that true? Think step by step and give me your reasoning. Then suggest how I can verify it."

 **Key Insight:** AI can **point you in the right direction**, but you must **verify** and **cite** real sources.

Learning with AI

AI is a powerful **tutor** when used correctly. It can explain concepts, give practice problems, and check your understanding but it should never do your homework for you.



Learning with AI

Learning tasks might include:

- Understanding new concepts in math, science, history, etc.
- Preparing for tests, quizzes, or exams
- Practicing skills like problem-solving or language
- Getting explanations that fit your level
- Any task where you want to deepen your understanding

How to prompt for learning:

Use **chain-of-thought** to see reasoning, **personas** as tutors, and **few-shot** to give examples of the kind of explanation you want. Always **try it yourself** after seeing the AI's help.

Prompting for Learning - Examples

Goal	Prompt (with techniques)
Math explanation	"Act as a patient math tutor . Show me step by step how to solve $3x + 5 = 20$. After explaining, give me a similar problem to solve on my own."
Science concept	"Explain how a solar eclipse happens to a 5th grader. Use an analogy with a flashlight and a ball. Keep it under 5 sentences."
Test preparation	"I'm studying for a test on the American Revolution. Give me 5 quiz questions about key events, with answers hidden (I'll check after). Format as a numbered list."
Language practice	"I'm learning Spanish. Give me a short dialogue in Spanish between two friends meeting at a park. Include an English translation after each line."

 **Key Insight:** AI helps you **understand**, then you **apply** what you learned.

Mini Activity 2 :

Prompting for Learning & Research

Instructions: For each scenario, write a prompt that would help you **learn** (not just get an answer) or **research** (gather and verify information). Use at least one advanced technique.

Scenario	Your Prompt
You need to understand the difference between a plant cell and an animal cell.	
You're writing a report on the water cycle and need reliable facts.	
You're stuck on a math problem: $\frac{2}{3} + \frac{1}{4}$.	
You're preparing for a debate about whether schools should start later.	

Sample answers on next slide.

Mini Activity 2 : Answer Key

Scenario	Sample Prompt
Plant vs animal cell (learning)	"Act as a biology tutor . Explain the differences between a plant cell and an animal cell in 3 bullet points. After your explanation, give me a one-sentence question to check my understanding."
Water cycle report (research)	"I need facts about the water cycle for a 5th grade report. List 3 reliable sources (book titles or websites) and summarize one key fact from each. If you're unsure, say so."
Math problem (learning)	"Show me step by step how to add $\frac{2}{3} + \frac{1}{4}$. After you explain, give me a similar fraction addition problem to solve on my own."
Debate preparation (research + reasoning)	"I'm preparing for a debate on later school start times. Give me two arguments for and two arguments against , each with a short explanation. Also suggest how I can verify the facts."



Writing with AI

AI can be a **writing partner** - helping you brainstorm, outline, draft, revise, and polish. But the final words should be **yours**. Use AI to **improve your writing**, not to write for you.

Writing with AI

Writing tasks might include:

- Essays, reports, or research papers
- Stories, poems, or scripts
- Letters, emails, or messages
- Speeches, presentations, or scripts
- Any task where you express ideas in text

How to prompt for writing:

Use **personas** (editor, coach), **few-shot** to show style, and **iterative prompting** to refine your draft. Always **write the first version yourself** if possible. Use **QuillBot** to paraphrase or polish your own writing.

Prompting for Writing - Examples

Goal	Prompt (with techniques)
Essay outline	"I'm writing an essay about why recycling is important. Help me create an outline with: introduction, 3 body paragraphs (each with a main idea), and conclusion. Use bullet points."
Story revision	"Here's the opening of my story: 'The old house stood at the end of the lane, its windows like eyes watching me.' Act as a writing coach and give me three suggestions to make it more suspenseful. Keep the original sentence intact."
Email draft	"Draft a polite email to my teacher asking for feedback on my project draft. Keep it short and respectful. Include that I will send the file separately."
Poetry style	"Here's an example of a short poem about the ocean: 'Waves crash on the shore / Salt spray in the air / Endless blue stretching far.' Now write a short poem about a thunderstorm in a similar style: three lines, nature theme."

 **Key Insight:** AI helps you **shape** your writing; you keep the **voice**. QuillBot can help you rephrase a sentence if you're stuck, but always keep your own ideas.

Responsible AI Use (The Golden Rules)

Using AI responsibly means using it as a **thinking partner**, not a shortcut. You stay in control, you verify the output, and you never submit AI work as your own.

Rule	What It Means	Example
You are the author	Never copy-paste AI output and call it your own.	 Use AI to outline, then write your own version.  Submit AI-generated essay directly.
Verify facts	AI can hallucinate. Always check important information.	 Use NotebookLM with your sources.  Trust ChatGPT without checking.
Stay in the loop	You decide, you review, you are responsible.	 Refine prompts and check outputs.  Accept the first answer blindly.
Use AI to learn, not to cheat	AI should help you understand, not do your homework.	 "Explain step by step, then give me a similar problem."  "Solve this problem for me."
Credit your tools	If your teacher or school asks, be honest about using AI.	 "I used AI to brainstorm ideas."  Hide your use of AI.

Why Responsible Use of AI Matters

Using AI responsibly builds trust, improves your own skills, and prepares you for a future where AI is everywhere.



Build Trust

Being honest about AI use builds credibility with teachers and peers.



Improve Your Skills

Using AI as a thinking partner sharpens your own reasoning and creativity.



Prepare for the Future

AI is everywhere – learning to use it responsibly now sets you up for success.

You are the human in the loop.

The skills you've built around prompting are powerful but they're only as good as how you use them. Use AI to **support your thinking**, not to **skip it**. Keep growing your **My Prompt Library** - it's your personal AI Prompt handbook for life.

✓ Spark ideas

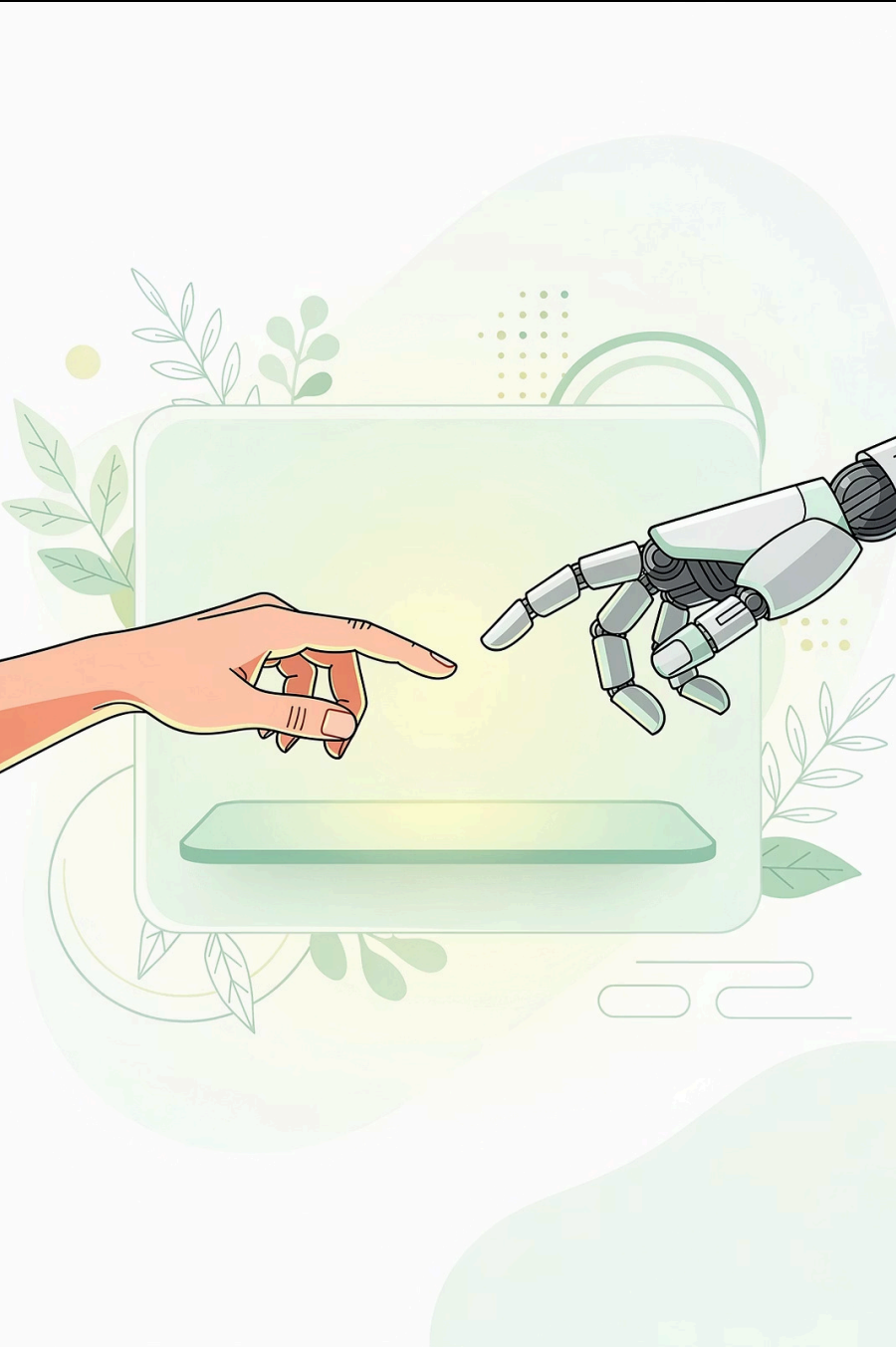
Let AI open doors you didn't know existed.

✓ Stay in control

You decide, you verify, you are responsible.

✓ Keep growing

Every prompt you write makes you a better thinker.



Hands-on Activity : Real-Task Prompt Playground

Goal: Apply your prompting skills to a **real task you actually have** - whatever category fits. Use the framework and advanced techniques, and document the process in your **My Prompt Library**.

- ❑ **Setup:** Use **ChatGPT, Gemini, NotebookLM, or QuillBot** as appropriate for your task. Open your **My Prompt Library** document.



Step 1: Choose a Real Task

Pick a task that is meaningful to you. It could be:

Category	Task Examples
Creative	Invent a new character, brainstorm a story idea, design a game concept.
Everyday	Plan a weekend, make a packing list, create a weekly schedule.
Research	Summarise a topic, compare two sources, find key facts.
Learning	Make a study guide, explain a concept, generate practice questions.
Writing	Draft a story opening, improve a sentence, check tone.

Steps 2 & 3 : Write and Iterate

Write a First Prompt

Using what you learned in S11 and S12, write a prompt that will help you **learn or create** (not just get an answer). Use at least one advanced technique (few-shot, role-based, chain-of-thought) if it fits.

Iterate

Run your prompt in **ChatGPT** or **Gemini**. Look at the output. What's good? What's missing? Refine your prompt at least once. Save both versions.

Steps 4, 5 & 6 : Verify, Polish, and Save

Step 4: Verify or Reflect

- **For Research/Learning:** Upload a trusted source (e.g., a textbook, Wikipedia, a reliable website) to **NotebookLM**. Ask the same questions. Compare with your draft. Correct any mistakes.
- **For Creative/Writing:** Ask yourself: *"Does this match my vision? What would I change?"*

Step 5: Polish with QuillBot

Copy your final draft into **QuillBot**. Select "Paraphrase" from Tools, Choose a mode (e.g., "Standard" or "Fluency") and review the suggestions. Accept changes that improve clarity or tone without changing your meaning.

Step 6: Add to Your Prompt Library

Create a **Prompt Library** entry for your best version. Use the flexible format from Session 11. Include: Date, Goal, AI Tools used, Final Prompt, Elements Used, Verification Steps, What Worked, What I'd Change Next Time, AI's Response (optional).

Prompt Library Entry Format

Use the flexible format from Session 11. Include the following fields:

Date

When you created the prompt

Goal

e.g., "Create a study guide for the water cycle"

AI Tools Used

ChatGPT/Gemini, NotebookLM, QuillBot

Final Prompt

The best version after refinement

Elements Used

Optional: which techniques you used

Verification Steps

What you checked and how

What Worked

Optional

What I'd Change

Optional: next time improvements

Summary

Key Term	What It Means
Creative tasks	Storytelling, art, games, imagination - AI sparks ideas, you create.
Everyday tasks	Planning, organizing, deciding - AI helps you think, you decide.
Research	Gathering facts, comparing viewpoints - AI summarizes, you verify and cite.
Learning	Understanding concepts, practicing skills - AI explains, you apply.
Writing	Drafting, revising, polishing - AI helps shape, you write.
Responsible AI use	Use AI as a thinking partner; never cheat; always verify.

Homework

Part 1: Quick Warm-Up

Match each task type to the best AI tool for the job. Write your answers.

Task	Best Tool (ChatGPT, Gemini, NotebookLM, QuillBot)
1. You need to summarise a chapter from your science textbook.	
2. You want to rewrite a sentence to sound more formal.	
3. You're brainstorming a magical creature for a story.	
4. You need to verify facts from a Wikipedia article.	

 **Answers:** 1 – NotebookLM, 2 – QuillBot, 3 – ChatGPT or Gemini, 4 – NotebookLM with the article as source

Homework

Part 2: Your Real-Task Prompt

Pick **one** real task from the list below (or use one you actually have this week). Write a prompt for that task using at least one advanced technique (few-shot, role-based, chain-of-thought). Run it in ChatGPT or Gemini. Write down the AI's response and one way you would refine it.

- **Creative:** Create a name and a short description for a new ice cream flavor.
- **Everyday:** Make a simple to-do list for a busy day (e.g., school, homework, chores).
- **Research:** Find three interesting facts about penguins (then verify with NotebookLM).
- **Learning:** Ask AI to explain how a caterpillar turns into a butterfly in three simple steps.
- **Writing:** Improve this sentence: "The movie was good but also had some parts that were not so good."

Homework

Part 3: Add to My Prompt Library

After completing your core task, add an entry to your **My Prompt Library** for the prompt you used. Use the flexible format (Date, Goal, Final Prompt, What Worked). This is part of your ongoing library - you will keep adding to it in future sessions.

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Next Session Preview

Session 14 : AI That Reads - Natural Language Processing (NLP)

Now that you've mastered prompting for real tasks, we'll explore the technology behind AI's ability to read and understand language. You'll learn about:

Text Analysis

How AI analyzes text (sentiment, summaries, etc)

Language Tools

Tools like QuillBot and translation apps

NLP in Action

How NLP powers the chatbots you've been using

📖 Get ready to dive into the "language brain" of AI!